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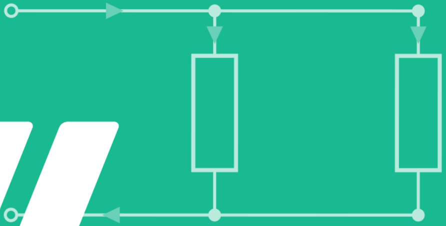
Modul 4: Basic DC networks



Stand: 18. Februar 2026

License

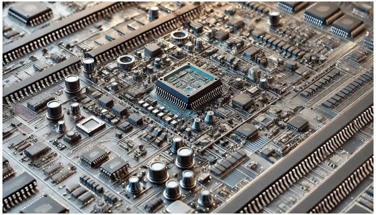
Electrical networks consist of:

- Two-pole or multi-pole: Elements with two or more electrical connections
- Connected by ideal conductive connections

These can be connected to systems of various sizes:

- Energy distribution networks ¿1000 km in length
- Electronic circuits in the micrometer range (microcontrollers)

Symbolbilder

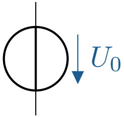
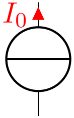
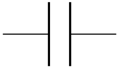


Electrical networks

Objective: Calculation of electrical quantities in a circuit

- Simplification of real components to ideal two-pole circuits
- Representation of two-pole circuits using equivalent circuit diagrams (see table)
  $\rightarrow$ form the basis for mathematical calculations
- Equivalent circuit diagrams model the basic behavior of real electrical circuits

Examples of two poles:

Dipole	Circuit symbol
Ideal voltage source	
ideal current source	
Capacitor (capacitance)	
ohmic resistance	
Coil (inductance)	

Learning objectives: Two-terminal elements and reference direction systems

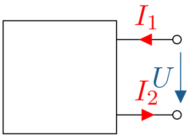
The students can

- explain the concept of two poles, name common two poles and use their circuit symbols
- explain the generator and consumer arrow system and apply it in accordance with conventions
- explain the central elements of the basic circuit and recognise the relationships between them

Two-terminal elements

Generally:

- Two external connections
- Clamping currents I_1 and I_2 are equal in magnitude ($I_1 = I_2 = I$)
- Clamping behaviour (ratio of I and U)
- Actual construction dimensions, material properties, and field strength distribution in the component are neglected.

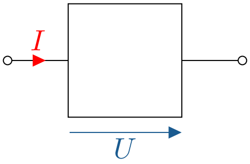
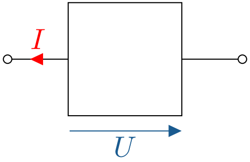


Active two-phase: contains current/voltage source(s) and, if necessary, resistors, capacitors, coils...
Passive two-pole: Contains no sources, only resistors, capacitors, coils...

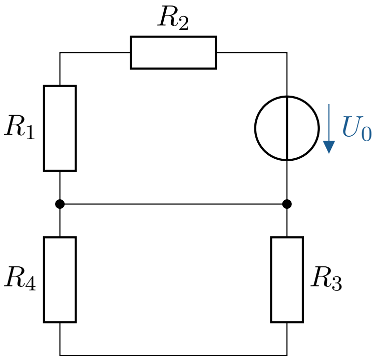
Sign convention

Zählpfeilrichtung theoretisch beliebig, jedoch Konventionen:

- Passive sign convention (PSC): Current and voltage **In the same way** (passive two-pole devices, e.g. resistors)
- Generator sign convention (GSC): Current and voltage **Opposite** (active two-pole devices, e.g. voltage sources)

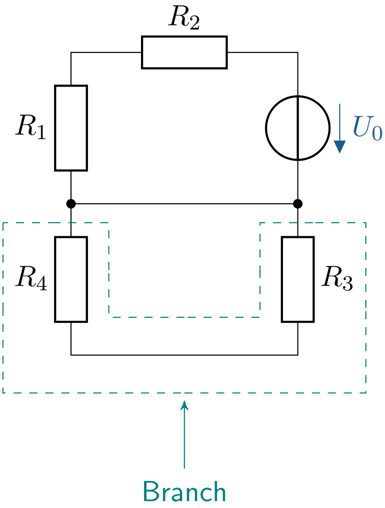
sign convention	Generated power	Consumed power	Passive sign-convention arrows
PSC	$P = -UI$	$P = UI > 0$	
ASC	$P = UI > 0$	$P = -UI$	

The most important terms are shown in the illustration:



The most important terms are shown in the illustration:

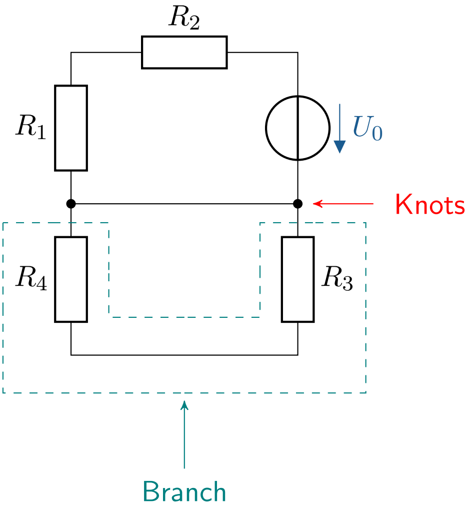
- idealised two-poles form **branches** of the network



The basic circuit

The most important terms are shown in the illustration:

- idealised two-poles form **branches** of the network
- Connection of the individual branches via **Knots** K_n



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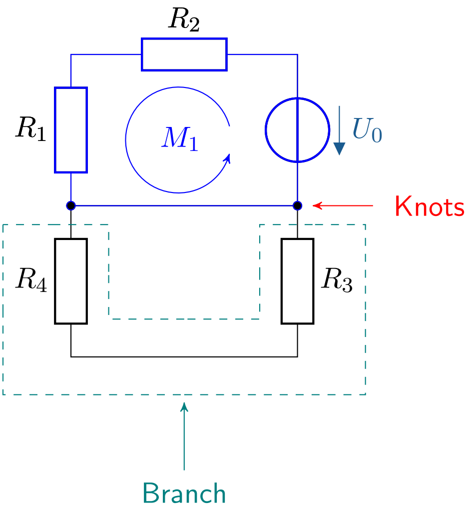
- Forms a closed chain of branches **Mesh** M_n



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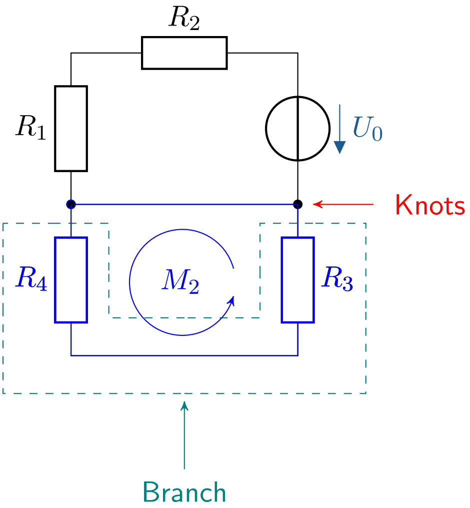

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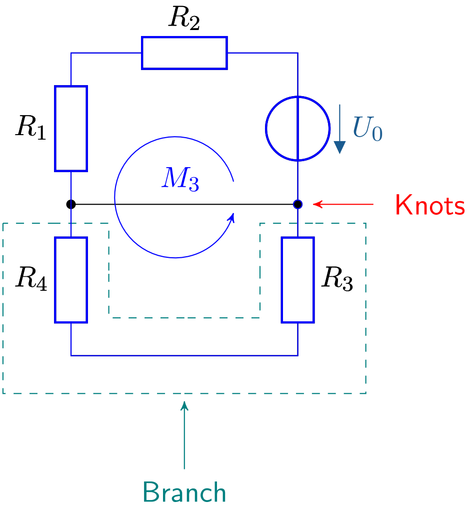

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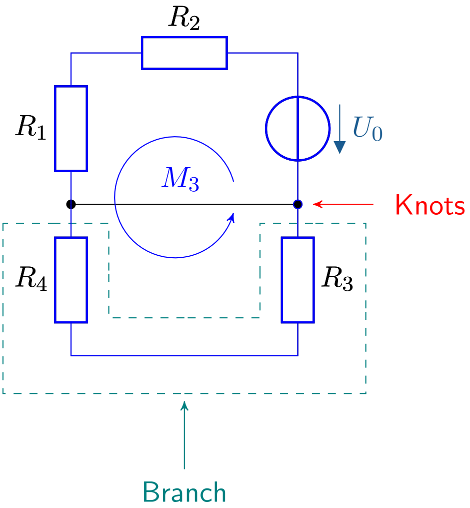
How are current and voltage distributed in a network?



Learning objectives: Kirchhoff's rules

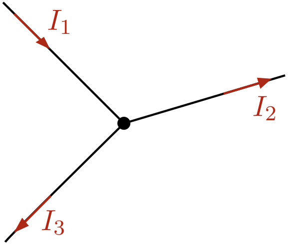
The students can

- reproduce the key statements of Kirchhoff's rules
- apply Kirchhoff's rules to simple resistance networks

Node rule I (Kirchhoff's 1st law)

Key point:

Total inflows
=
Total outflows



$$\sum I_{\text{in}} = \sum I_{\text{out}}$$

$$I_1 = I_2 + I_3$$

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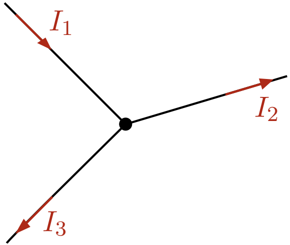


$$\sum I_{\text{in}} = \sum I_{\text{out}}$$

$$I_1 = I_2 + I_3$$

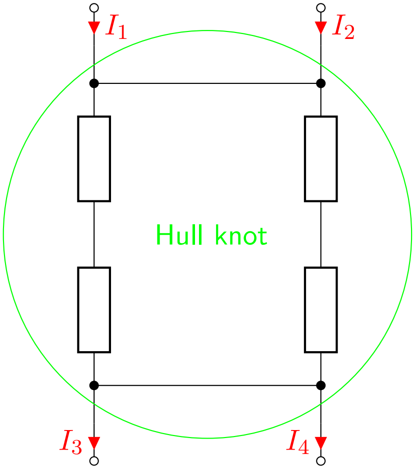
Alternatively: count incoming currents as positive and outgoing currents as negative:

$$I_1 - I_2 - I_3 = 0$$

$$\sum_{k=1}^{n} I_k = 0$$

Knot rule II

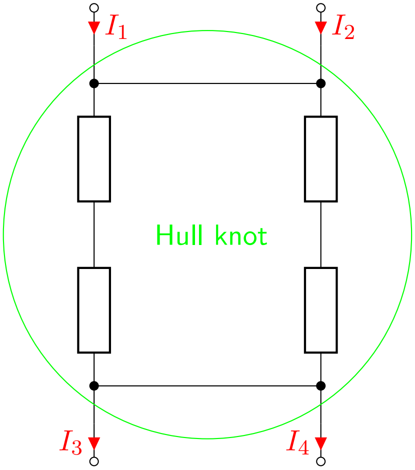
Application to envelope nodes:



$$I_1 + I_2 - I_3 - I_4 = 0$$

Knot rule II

Application to envelope nodes:



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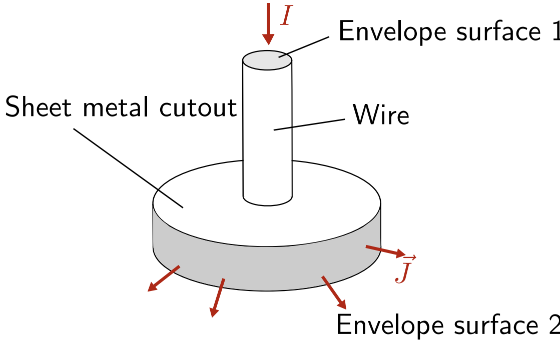
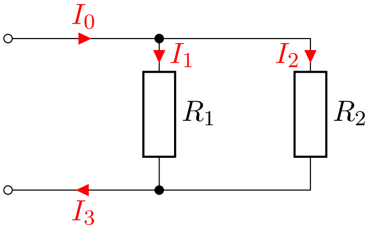
Application to geometric structures:



$$\iint\limits_A \vec{J} \cdot d\vec{A} = 0$$

Example: Parallel connection of resistors

In a parallel circuit, the currents split at the common node. How large is the current I_3?



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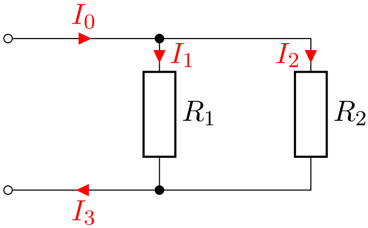
$I_0 - I_1 - I_2 = 0$
$I_0 = I_1 + I_2$
$I_1 + I_2 = I_3$
$\rightarrow I_3 = I_0$



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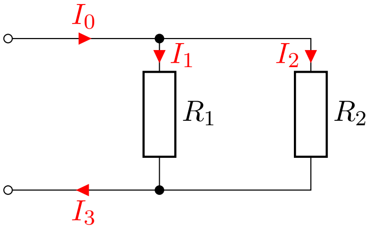

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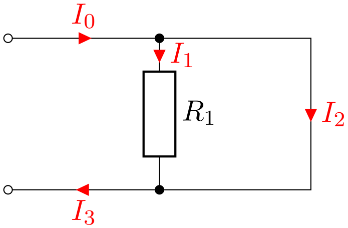
A resistor is replaced by a conductive connection. How strong is the current? I_2?



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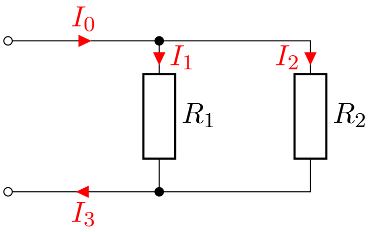

$$
\begin{aligned}
& I_0 - I_1 - I_2 = 0 \\
& I_0 = I_1 + I_2 \\
& I_1 + I_2 = I_3 \\
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\end{aligned}
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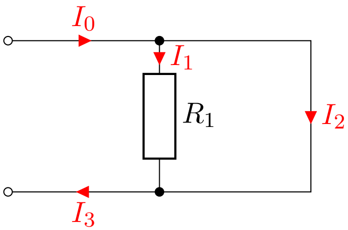


$$
\begin{aligned}
& I_0 = I_1 + I_2 \\
& U_2 = R_2 \cdot I_2 = 0 \cdot I_2 \\
& U_1 = R_1 \cdot I_1 \stackrel{!}{=} U_2 = 0 \\
& \rightarrow I_1 = 0 \\
& \rightarrow I_2 = I_0 = I_3
\end{aligned}
$$

Mesh rule (Kirchhoff's second law)

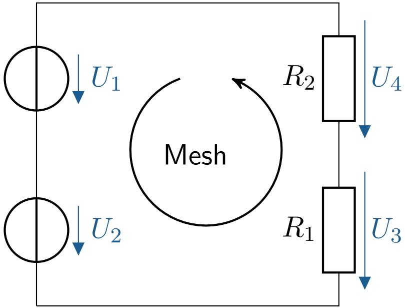
With closed mesh circulation:

Key point:

Sum of voltage sources

$=$

Total voltage at consumers



$$U_1 + U_2 = U_3 + U_4$$

Mesh rule (Kirchhoff's second law)

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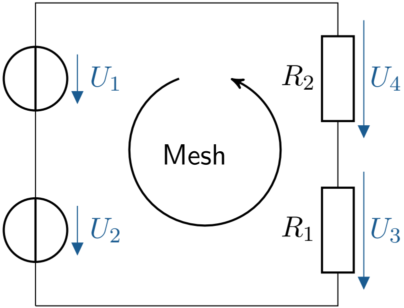

Key point:

Sum of voltage sources
=
Total voltage at consumers

Or:
With the mesh circulation closed, add up the partial stresses with the correct sign.

$$U_1 + U_2 = U_3 + U_4$$
$$U_1 + U_2 - U_3 - U_4 = 0$$

Merke:

$\sum_{k=1}^{n} U_k = 0$

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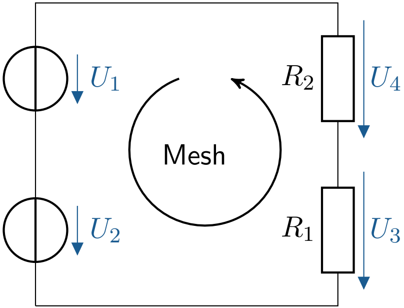

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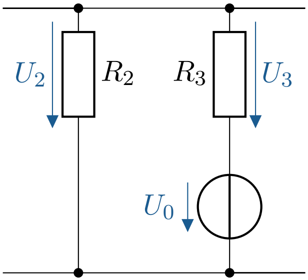
General description according to Maxwell's equations:

$$\oint_s \vec{E} d\vec{s} = 0$$

Example: Series connection of resistors

In a series connection, the voltages add up to a total voltage.
How high is the voltage? U_0?



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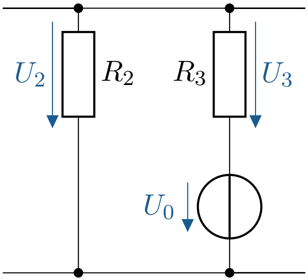

$U_2 - U_0 - U_3 = 0$
$\rightarrow U_0 = U_2 - U_3$

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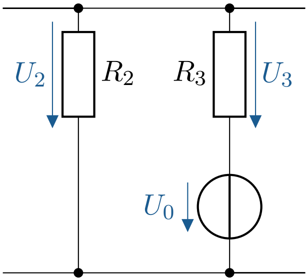

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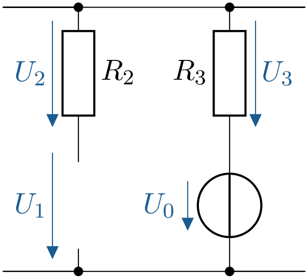
A connection is interrupted. How large is the voltage U_1 at the point of interruption?



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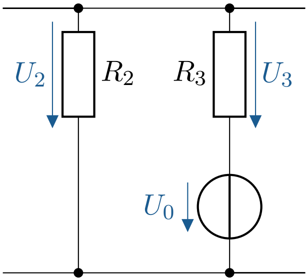

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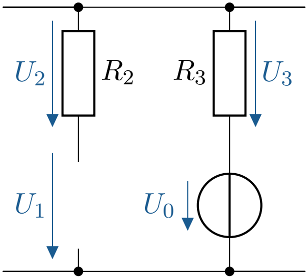


$$U_2 + U_1 - U_0 - U_3 = 0$$
$$U_2 = I_2 \cdot R_2$$
$$\rightarrow I_2 = 0$$
$$\rightarrow U_2 = 0$$
$$U_1 = U_0 + U_3$$

Learning objectives: Simple direct current networks

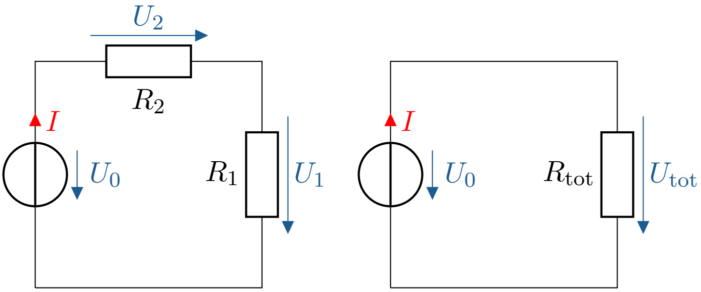
Learning objectives: Simple resistance networks

The students can

- Identify subcircuits in direct current networks
- Simplify and summarise resistance networks
- Determine short-circuit and no-load data
- Apply superposition methods

Mesh rule:
$U_0 - U_1 - U_2 = 0$

Ohm's law:
$U_1 = R_1 \cdot I \qquad U_2 = R_2 \cdot I$



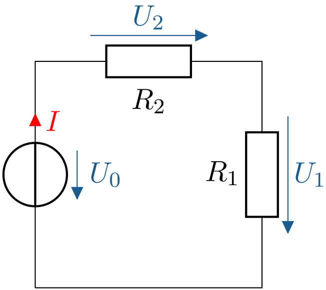
$U_0 - R_{\text{tot}} \cdot I = 0$
$R_{\text{tot}} = R_1 + R_2$

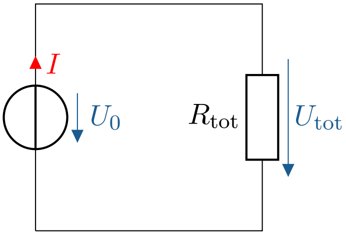
Mesh rule:
$U_0 - U_1 - U_2 = 0$

Ohm's law:
$U_1 = R_1 \cdot I \qquad U_2 = R_2 \cdot I$

The current I flows through all components:
$U_0 - R_1 \cdot I - R_2 \cdot I = 0$
$U_0 - (R_1 + R_2) \cdot I = 0$

Total resistance of a series connection of resistors:





$U_0 - R_{tot} \cdot I = 0$
$R_{tot} = R_1 + R_2$

Key point:

$R_{tot} = \sum_{k=1}^{n} R_k$

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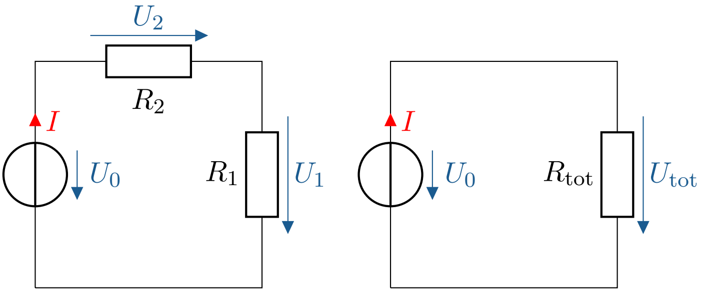


$U_0 - R_{tot} \cdot I = 0$
$R_{tot} = R_1 + R_2$

Total resistance of a series connection of resistors:

Key point:

$$R_{tot} = \sum_{k=1}^{n} R_k$$

Total conductance of a series connection of resistors:

Key point:

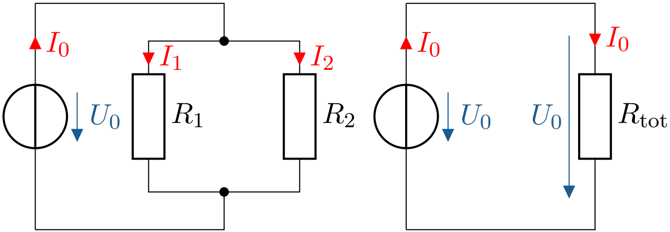
$$\frac{1}{G_{tot}} = \sum_{k=1}^{n} \frac{1}{G_k}$$

Parallel connection of resistors

Mesh rule: $U_{\mathrm{R1}} = U_{\mathrm{R2}} = U_0$
Knot rule: $I_0 - I_1 - I_2 = 0$
Ohm's law: $I_1 = \frac{U_0}{R_1}$

$$I_0 - \frac{U_0}{R_1} - \frac{U_0}{R_2} = 0$$

$$I_0 - \left(\frac{1}{R_1} + \frac{1}{R_2}\right) \cdot U_0 = 0$$



Key point:

$$G_{\mathrm{tot}} = \sum_{k=1}^{n} G_k$$

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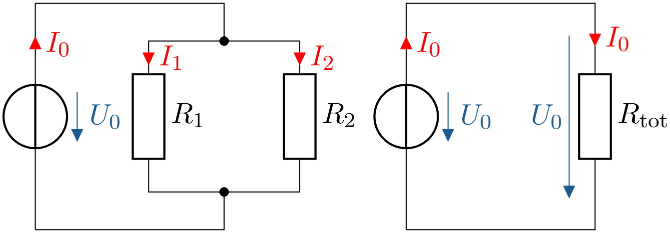


Key point:

$$\frac{1}{R_{\mathrm{tot}}} = \sum_{k=1}^{n} \frac{1}{R_k}$$

Key point:

$$G_{\mathrm{tot}} = \sum_{k=1}^{n} G_k$$

Voltage dividers in resistor networks

Series connection divides voltage U_0,
Current identical in all components

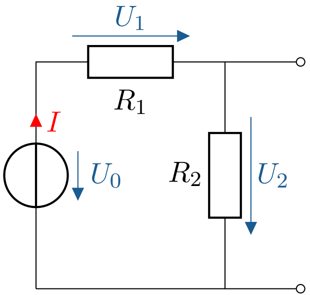
$$I = \frac{U_1}{R_1} = \frac{U_2}{R_2}$$

Series connection of resistors:

$$I = \frac{U_0}{R_1 + R_2}$$

Relationship between total voltage U_0 and partial voltage U_2:

$$\frac{U_2}{R_2} = \frac{U_0}{R_1 + R_2}$$



Key point:

$$U_2 = U_0 \cdot \frac{R_2}{R_1+R_2}$$

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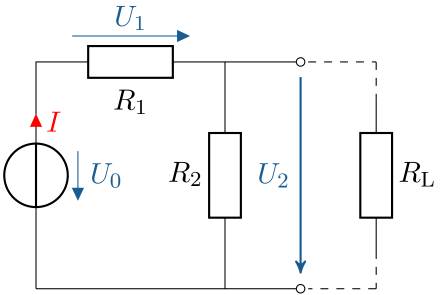


Key point:

$U_2 = U_0 \cdot \frac{R_2}{R_1+R_2}$

Only applies to **unloaded voltage dividers** (load current $= 0$)!
Load case: Combine R_2 and R_L!

Current dividers in resistor networks

Parallel connection divides current I_0, voltage identical across all components

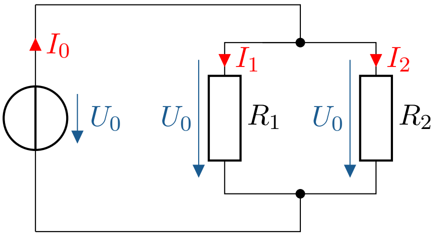
$$U_0 = I_1 \cdot R_1 = I_2 \cdot R_2$$

Parallel connection of resistors:

$$U_0 = I_0 \cdot R_{\text{tot}}$$

$$I_2 \cdot R_2 = I_0 \cdot R_{\text{tot}}$$

$$I_2 \cdot R_2 = I_0 \cdot \frac{R_1 \cdot R_2}{R_1 + R_2}$$



$$R_{\text{tot}} = \frac{R_1 \cdot R_2}{R_1 + R_2}$$

Key point:

$$I_2 = I_0 \cdot \frac{R_1}{R_1 + R_2}$$

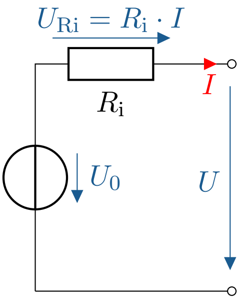
The real source of tension

Possesses internal resistance R_i

Unloaded real voltage source:

$$I = 0 \rightarrow U_\mathrm{Ri} = 0$$

$$U = U_0$$



Possesses internal resistance R_i

Unloaded real voltage source:

$$I = 0 \rightarrow U_\mathrm{Ri} = 0$$

$$U = U_0$$

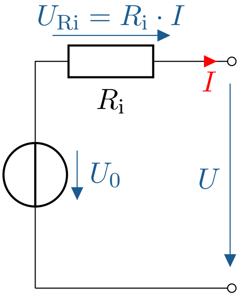
Loaded real voltage source:

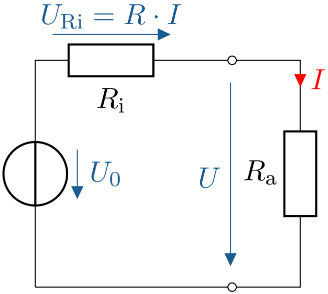
$$I > 0 \rightarrow U_\mathrm{Ri} > 0$$

$$U = U_0 - U_\mathrm{Ri} < U_0$$

Internal resistance as low as possible:

$$U_0 >> I \cdot R_\mathrm{i}$$





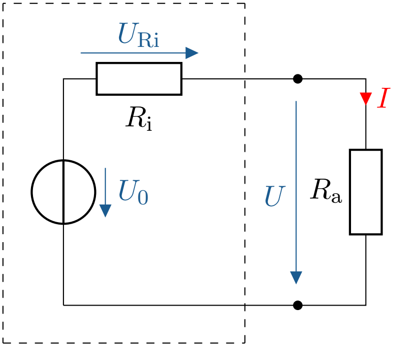
The replacement voltage source

Each **active** two-pole can be represented as a **substitute voltage source**

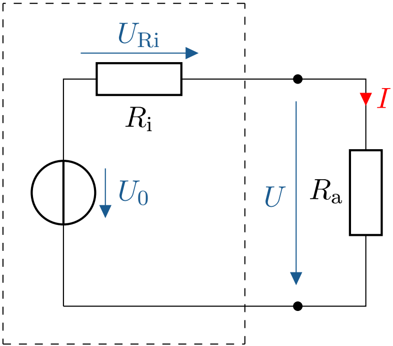
Ideal voltage source U_0 and **Internal resistance** R_i

$$U_0 = U_{\mathrm{Ri}} + U$$

$$U_0 = I \cdot R_{\mathrm{i}} + U$$



Each **active** two-pole can be represented as a **substitute voltage source**



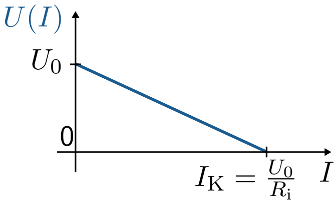
Ideal voltage source U_0 and **Internal resistance** R_i

$$U_0 = U_{\mathrm{Ri}} + U$$

$$U_0 = I \cdot R_{\mathrm{i}} + U$$

Reduction of voltage at the output:

$$U = U_0 - I \cdot R_{\mathrm{i}}$$

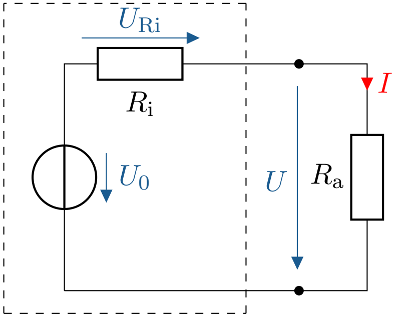


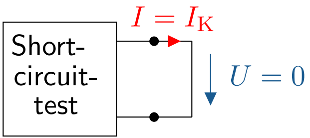
Determination of U_0 and R_i via **short circuit** and **no-load test**:

Short circuit: $R_a = 0$, $I = I_\mathrm{K}$ = short-circuit current

$$U = 0 \rightarrow U_0 = U_\mathrm{Ri}$$

$$U_\mathrm{Ri} = R_\mathrm{i} \cdot I_\mathrm{K} \rightarrow I_\mathrm{K} = \frac{U_0}{R_\mathrm{i}}$$





Determination of U_0 and R_i via **short circuit** and **no-load test**:

Short circuit: $R_a = 0$, $I = I_\mathrm{K} =$ short-circuit current

$$U = 0 \rightarrow U_0 = U_\mathrm{Ri}$$

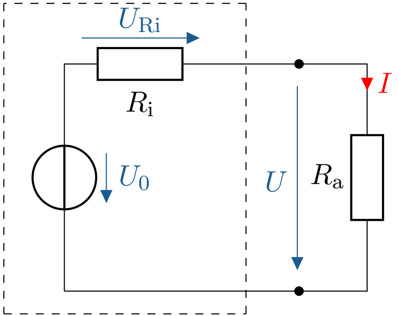
$$U_\mathrm{Ri} = R_\mathrm{i} \cdot I_\mathrm{K} \rightarrow I_\mathrm{K} = \frac{U_0}{R_\mathrm{i}}$$

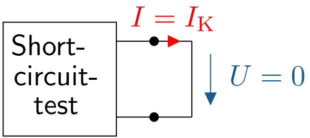
No load: $R_\mathrm{a} \rightarrow \infty$, $U = U_\mathrm{L} =$ no-load voltage

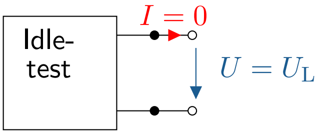
$$I = 0 \rightarrow U_\mathrm{Ri} = R_\mathrm{i} \cdot 0 = 0$$
$$U_\mathrm{L} = U_0$$

Key point:

$$U_0 = U_\mathrm{L}, \quad R_\mathrm{i} = \frac{U_\mathrm{L}}{I_\mathrm{K}}$$







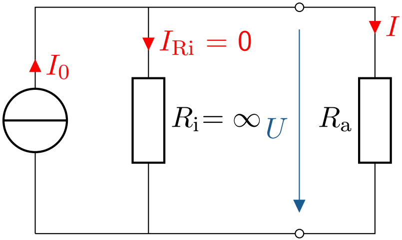
The real current source

Ideale Current source:
Internal resistance infinite:

$$I_{\mathrm{Ri}} = 0$$

$$I = I_0$$

Ideal current source



Ideale Current source:
Internal resistance infinite:

$$I_{\mathrm{Ri}} = 0$$

$$I = I_0$$

Reale Current source:
Possesses internal resistance $R_\mathrm{i} < \infty\,\Omega$

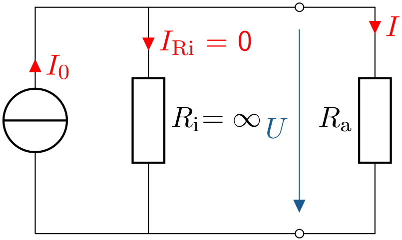
$$I_{\mathrm{Ri}} > 0$$

$$I = I_0 - I_{\mathrm{Ri}} < I_0$$

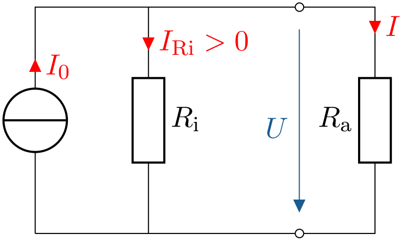
Internal resistance as high as possible:

$$I_0 >> U/R_\mathrm{i}$$

Ideal current source



Real current source



$I_{\mathrm{Ri}} = U/R_\mathrm{i}$

Each **active** two-pole can be represented as a **replacement current source**

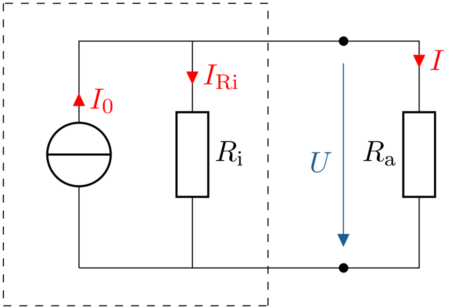
Iideal current source I_0 with **internal resistance** R_i

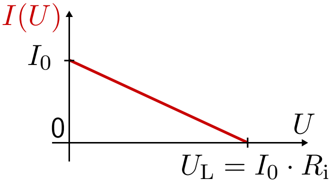
$$I_0 = I_{\mathrm{Ri}} + I$$

$$I_0 = \frac{U}{R_{\mathrm{i}}} + I$$

Reduction of output current depending on U:

$$I = I_0 - \frac{U}{R_{\mathrm{i}}}$$



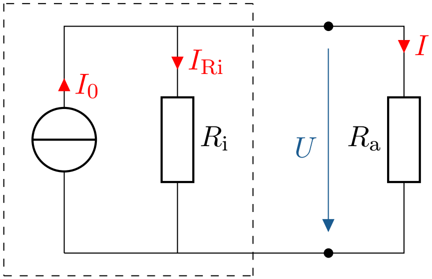


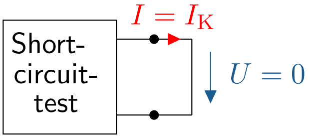
Determination of I_0 and R_i via **short circuit** and **no-load test**:

Short circuit: $R_a = 0$, $I = I_\mathrm{K}$ = short-circuit current

$$U = 0 \rightarrow I_\mathrm{Ri} = 0$$

$$I_\mathrm{K} = I_0$$





Determination of I_0 and R_i via **short circuit** and **no-load test**:

Short circuit: $R_a = 0$, $I = I_\mathrm{K}$ = short-circuit current

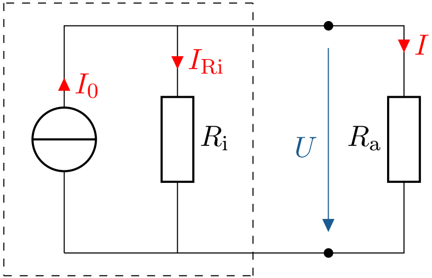
$$U = 0 \rightarrow I_\mathrm{Ri} = 0$$

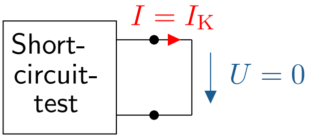
$$I_\mathrm{K} = I_0$$

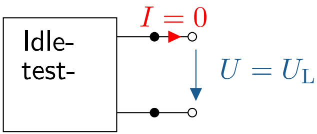
Idling: $R_\mathrm{a} \rightarrow \infty$, $U = U_\mathrm{L}$ = Open-circuit voltage

$$I = 0 \rightarrow I_\mathrm{Ri} = I_0$$

$$U_\mathrm{L} = R_\mathrm{i} \cdot I_0$$







Key point:

$$I_0 = I_\mathrm{K}, \quad R_\mathrm{i} = \frac{U_\mathrm{L}}{I_\mathrm{K}}$$

Internal resistance R_i remains identical, Circuit changes

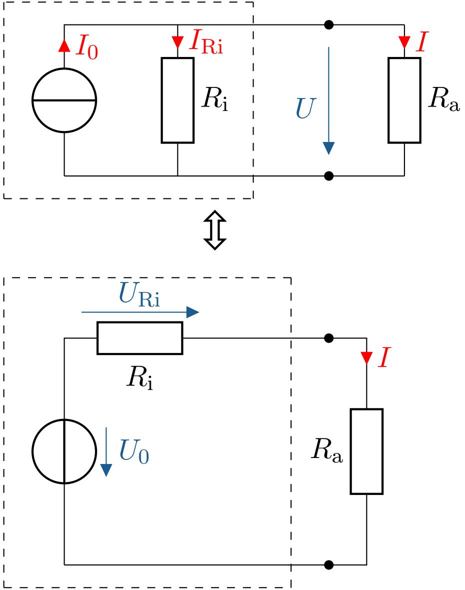
Convert **open-circuit voltage** and **short-circuit current** to another source:
Current source in voltage source:

$$U_0 = I_0 \cdot R_\mathrm{i}$$

Voltage source in current source:

$$I_0 = \frac{U_0}{R_\mathrm{i}}$$

The sign Conventions of the sources point in opposite directions!



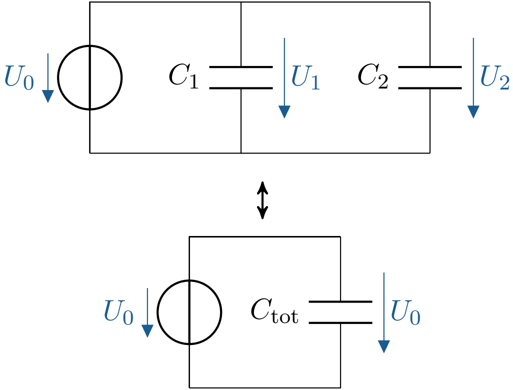
Parallel connection of capacities

Parallel connection: Voltage of all capacitances identical:

$$U_0 = U_1 = U_2$$

Charge per capacitor:

$$Q_k = C_k \cdot U_k, \quad Q_{\text{ges}} = C_{\text{ges}} \cdot U_0$$



Parallel connection of capacities

Parallel connection: Voltage of all capacitances identical:

$$U_0 = U_1 = U_2$$

Charge per capacitor:

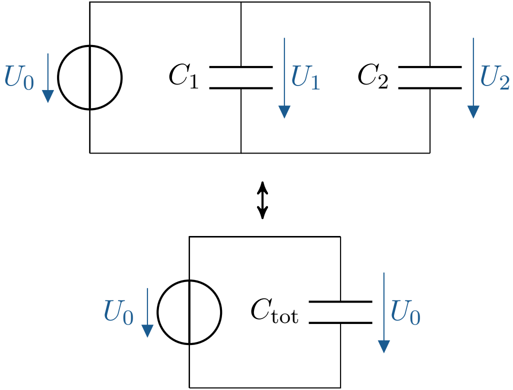
$$Q_k = C_k \cdot U_k, \quad Q_{\text{ges}} = C_{\text{ges}} \cdot U_0$$

Total load Q_{ges}:

$$Q_{\text{ges}} = Q_1 + Q_2 = C_1 \cdot U_0 + C_2 \cdot U_0$$

$$Q_{\text{ges}} = (C_1 + C_2) \cdot U_0$$

$$\rightarrow C_1 + C_2 = C_{\text{ges}}$$



Key point:

$$C_{\text{tot}} = \sum_{k=1}^{n} C_k$$

Series connection: all capacitors have the **same charge** Q!

Basic capacitor equation:

$$Q = C \cdot U$$

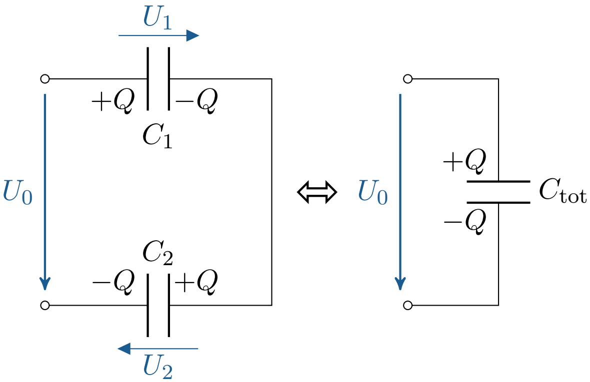
$$U_1 = \frac{Q}{C_1}, \quad U_2 = \frac{Q}{C_2}$$

Series connection of capacitances:

$$U_0 = U_1 + U_2 = \left(\frac{1}{C_1} + \frac{1}{C_2}\right) \cdot Q$$

Equivalent circuit diagram:

$$U_0 = \frac{1}{C_{\text{tot}}} \cdot Q$$



Key point:

$$\frac{1}{C_{\text{tot}}} = \sum_{k=1}^{n} \frac{1}{C_k}$$

Voltage divider on capacitances

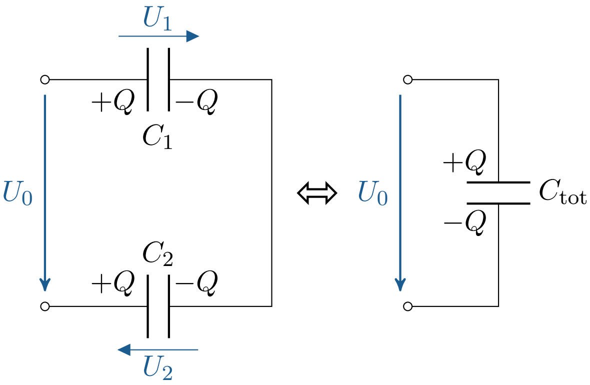
$$Q_1 = Q_2 = Q_{\text{tot}} = Q$$

$$C_{\text{tot}} = \frac{C_1 \cdot C_2}{C_1 + C_2}$$

$$U_0 = \frac{1}{C_{\text{tot}}} \cdot Q \to Q = C_{\text{tot}} \cdot U_0$$

$$U_1 = \frac{Q}{C_1} \to U_1 = \frac{C_{\text{tot}}}{C_1} \cdot U_0$$

$$U_2 = \frac{Q}{C_2} \to U_2 = \frac{C_{\text{tot}}}{C_2} \cdot U_0$$

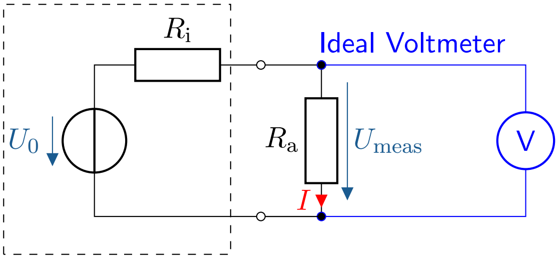


Key point:

$$U_1 = \frac{C_2}{C_1 + C_2} \cdot U_0, \quad U_2 = \frac{C_1}{C_1 + C_2} \cdot U_0$$

Voltage measurement in a direct current circuit:

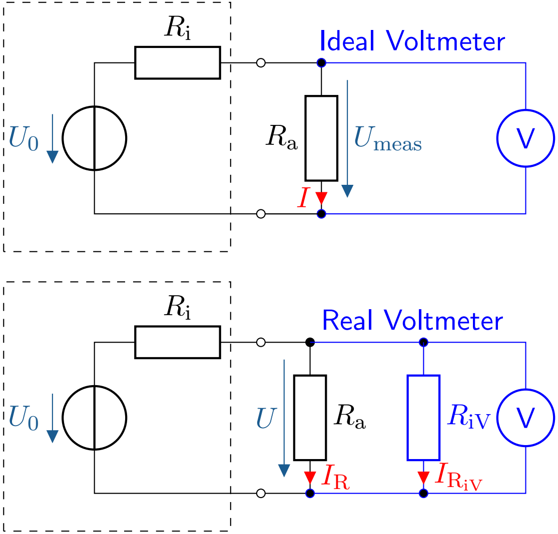
- Measurement with a **voltmeter**
- Measuring device **parallel** to the component (no interruption of the circuit necessary)



Voltage measurement in a direct current circuit:

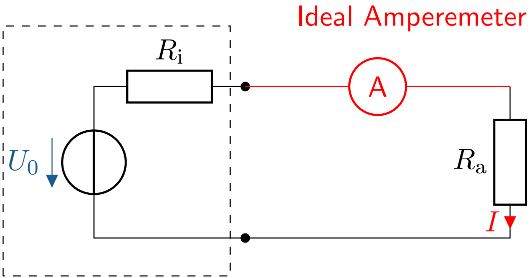
- Measurement with a **voltmeter**
- Measuring device **parallel** to the component (no interruption of the circuit necessary)
- Real voltmeter: **Internal resistance** $R_{\mathrm{iV}} >> R_{\mathrm{mess}}$
- measured voltage corrected for internal resistance U_{korr}:

$$U_{\mathrm{korr}} = U \cdot \left(1 + \frac{R_{\mathrm{i}}||R_{\mathrm{a}}}{R_{\mathrm{iV}}}\right)$$



Current measurement in a direct current circuit:

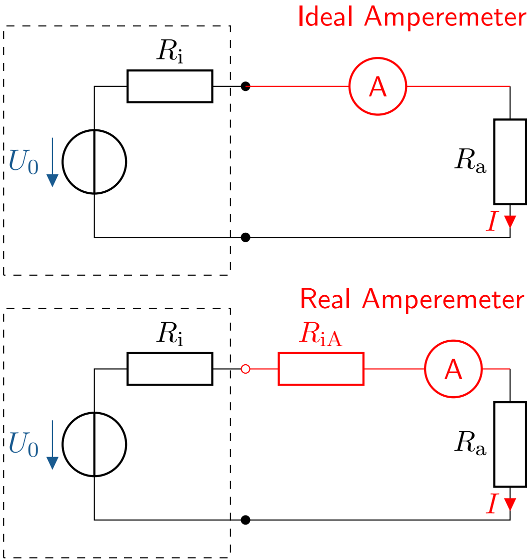
- Measurement with **ammeter**
- Measuring device **in series** with the component (disconnect the circuit!)



Current measurement in a direct current circuit:

- Measurement with **ammeter**
- Measuring device **in series** with the component (disconnect the circuit!)
- Real ammeter: Internal resistance $R_{iA} << R_a$
- Measured current I_{corr} corrected for internal resistance R_{iA}:

$$I_{korr} = I \cdot \left(1 + \frac{R_{iA}}{R_i + R_a}\right)$$



Correct measurement of current and voltage I

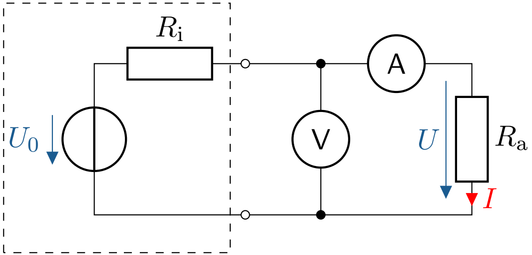
When measuring current and voltage **simultaneously**:

- mutual **influence** of the internal resistances **of the measuring instruments**

Current-correct Measurement:

- greater focus on current strength
- $U_{\mathrm{meas}} = U_{\mathrm{Ra}} + U_{\mathrm{Amperemeter}}$
- use when the following applies:

$$R_{\mathrm{iA}} \ll R_{\mathrm{a}}$$



Correct measurement of current and voltage II

Correct voltage measurement:

- Greater focus on excitement
- $I_{\text{meas}} = I_{\text{Ra}} + I_{\text{Voltmeter}}$
- Use when applicable:

$$R_{\text{iV}} \gg R_{\text{a}}$$

In unclear circumstances:

- correct current measurement for $R_{\text{a}} > \sqrt{R_{\text{iA}} \cdot R_{\text{iV}}}$
- correct voltage measurement for $R_{\text{a}} < \sqrt{R_{\text{iA}} \cdot R_{\text{iV}}}$